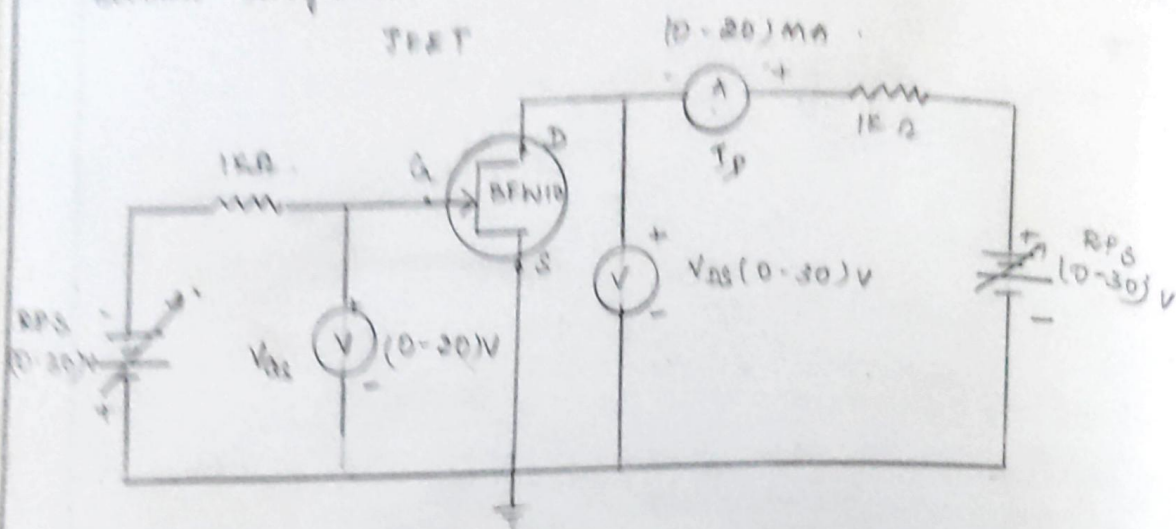
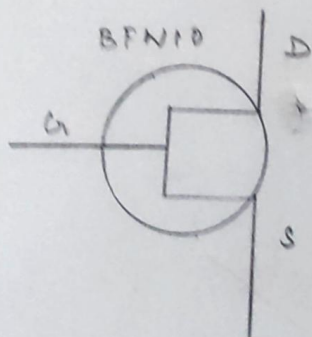


circuit diagram:



symbol



AIM:

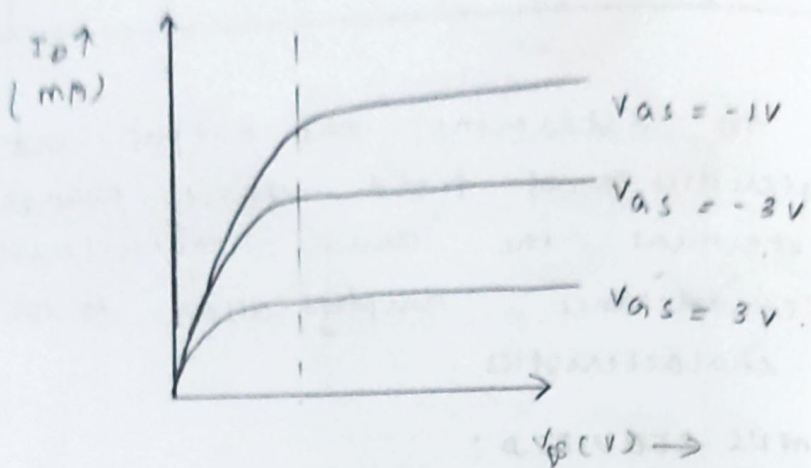
TO determine the output and transfer characteristics of field effect transistor and to determine the drain characteristics, trans conductance, amplification factor from the characteristics.

APPARATUS REQUIRED:

SL NO.	APPARATUS / COMPONENT NAME	TYPE / RANGE	QUANTITY
1.	Dual Regulated power supply (RPS)	(0-30)V	1
2.	Voltmeter	(0-10)V	1
3.	Ammeter	(0-20) mA	1
4.	Bread Board	-	1
5.	Junction field effect transistor (JFET)	BFW10	1
6.	point contact diode	0A79	1
7.	connecting wires	-	As required.

Model graph:

Active Region



TABULATION:

JFET

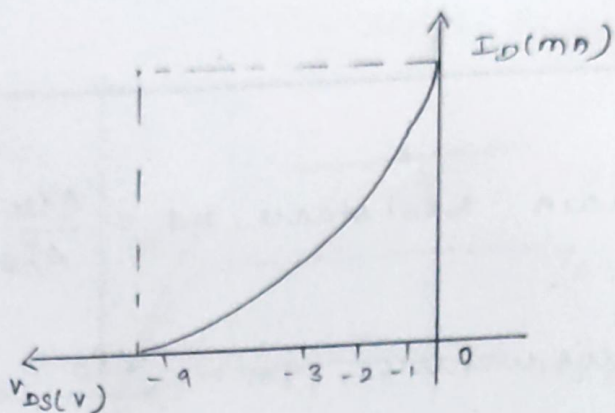
$V_{GS} = -1V$		$V_{GS} = -2V$		$V_{GS} = -3V$	
$V_{DS}(V)$	$I_D(mA)$	$V_{DS}(V)$	$I_D(mA)$	$V_{DS}(V)$	$I_D(mA)$
0.09	0.88	0.16	0.52	0.12	0.38
0.11	1.59	0.24	0.66	0.16	0.53
0.16	2.49	0.25	0.72	0.18	0.63
0.18	3.20	0.26	1.32	0.19	0.68
0.19	3.99	0.27	1.68	0.21	0.72
0.20	4.69	0.27	2.0	0.25	0.78
0.22	5.34	0.28	2.50	0.26	0.80
0.221	5.83	0.30	2.79	0.29	0.85
0.25	6.24	0.32	3.04	0.30	0.87
0.27	6.44	0.48	3.76	0.31	0.91
0.28	6.48	0.50	3.27	0.33	0.92
0.38	6.59	0.58	3.34	0.34	0.93
0.32	6.61	0.59	3.41	0.35	0.95
0.33	6.63	0.68	3.47	0.40	0.95
0.35	6.43	0.72			

FORMULA :

1. AC drain resistance, $r_d = \frac{\Delta V_{DS}}{\Delta I_D}$ at constant ΔV_{GS} .
2. Transconductance, $g_m = \frac{\Delta I_D}{\Delta V_{GS}}$ at constant ΔV_{DS} .
3. Amplification factor $\mu = r_d * g_m$.

THEORY :

TRANSFER CHARACTERISTICS



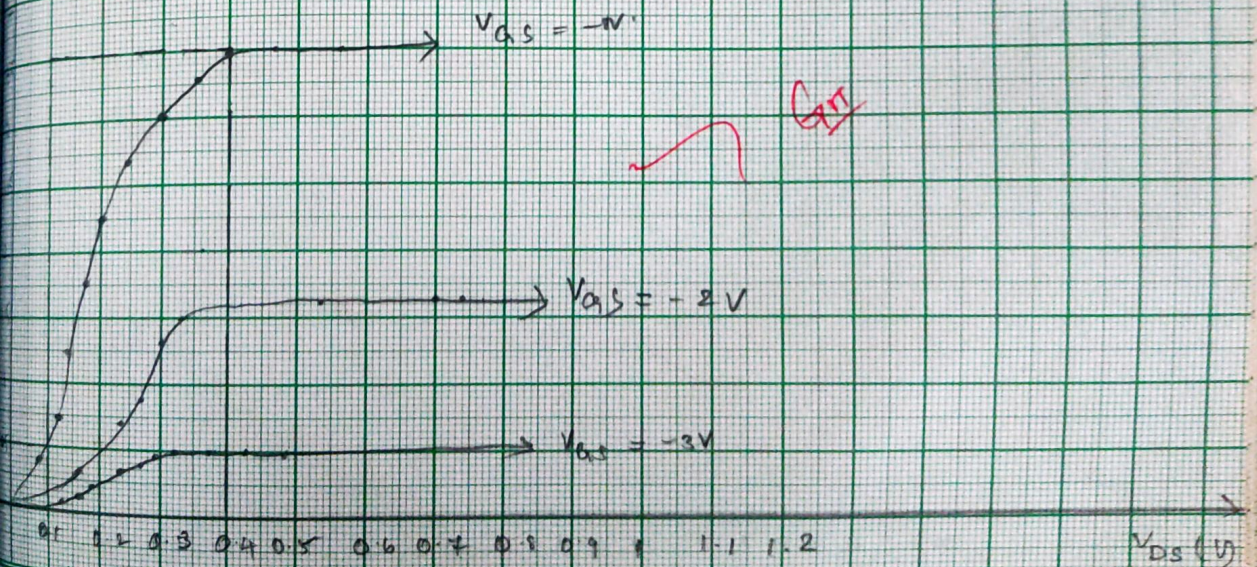
TABULATION:

$V_{DS} = 1V$		$V_{DS} = 3V$		$V_{DS} = 5V$	
$V_{GS}(V)$	$I_D(mA)$	$V_{GS}(V)$	$I_D(mA)$	$V_{GS}(V)$	$I_D(mA)$
0	4.97	0	7.68	0	8.81
-1.06	4.78	-1.01	6.84	-0.91	7.32
-1.33	4.59	-1.37	5.79	-1.32	6.20
-1.77	3.99	-1.45	3.39	-1.93	4.02
-2.42	2.26	-1.59	1.90	-2.54	2.11
-3.08	0.88	-2.69	0.98	-2.87	1.26
-3.54	0.04	-3.43	0	-3.53	0.09
-5.11	0	-4.13	0	-4.02	0
-4.47	0	-4.9	0	-4.55	0
-5.09	0			-5.11	0

Characteristics of JFET

Drain Characteristics

Scale
On x axis $1 \text{ cm} = 0.1 \text{ V}$
On y axis $1 \text{ cm} = 0.5 \text{ mA}$



$$r_d = \frac{\Delta V_{DS}}{\Delta I_D}$$

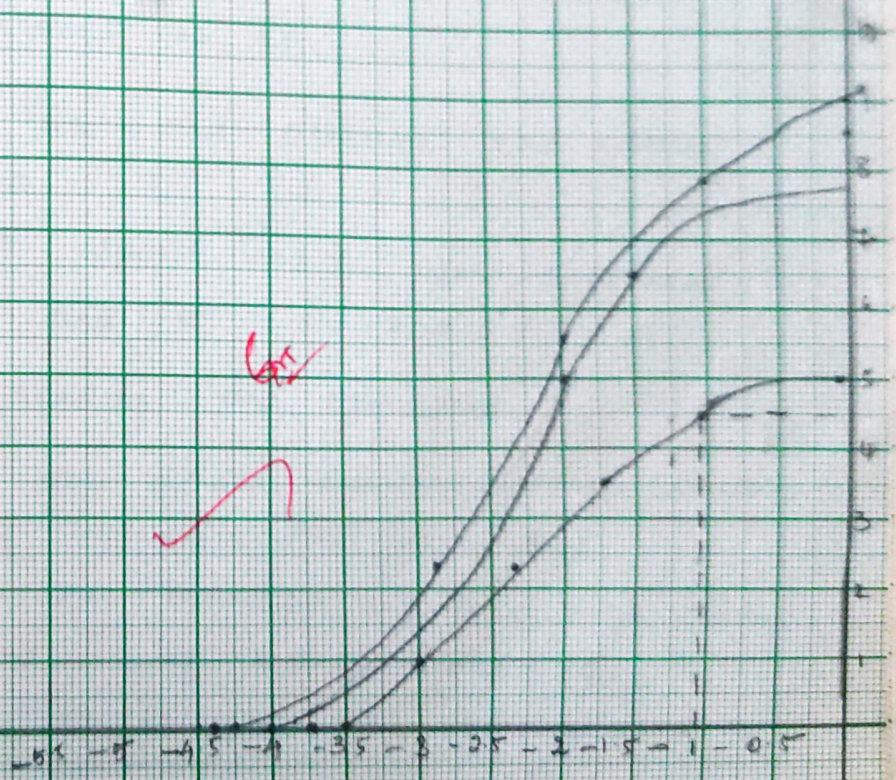
$$= \frac{6.1 - 0.3 \text{ V}}{1.2 - 6.4 \text{ mA}}$$

$$= 51.92 \times 10^{-3} \Omega$$

DATE

THEORY: 100% = 100%

LABORATORY: 100% = 100%



$$g_m = \frac{\Delta I_D}{\Delta V_{GS}}$$

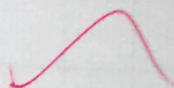
$$= \frac{5 - 4.6}{1.2 - 0}$$

$$= 333.3 \times 10^{-3} \Omega$$

Amplification factor $\mu = g_m \times r_d$

$$= 51.92 \times 10^{-5} \times 333.3 \times 10^{-3} \Omega$$

$$= 17.30 \times 10^{-3} \Omega$$



PROCEDURE

- * All the connections are made as per the circuit diagram.
- * To plot the drain characteristics keep V_{GS} constant at 0 V.
- * Vary the V_{DS} and observe the values of V_{DS} and I_D .
- * Repeat the above steps 2, 3 for different values of V_{GS} at 0.1 V and 0.2 V.
- * All the readings are tabulated.
- * To plot the transfer characteristics, keep V_{DS} constant at 1 V.
- * Vary V_{GS} and observe the values of V_{GS} and I_D .
- * Repeat step 6 and 7 for different values of V_{DS} and at 1.5 V and 2 V.
- * The readings are tabulated.
- * From the drain characteristics, calculate the values of dynamic resistance (r_d).
- * From the transfer characteristics, calculate the value of transconductance (g_m).
- * And also calculate the amplification factor (μ).



RESULT :

Superm

Thus the output and transfer characteristics of Junction field transistor (JFET) are determined. The following parameters are determined from the graph.

→ Ac drain resistance, $r_d = 51.92 \times 10^3 \Omega$

→ Transconductance, $g_m = 333.3 \times 10^3 \Omega$

→ Amplification factor, $\mu = 17.30 \times 10^3 \Omega$